ELEC60003
EE9-AC1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART III/IV: MEng, BEng.and ACGI

ANALOGUE INTEGRATED CIRCUITS AND SYSTEMS

Tuesday, 3 December 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : C. Toumazou
Second Marker(s) : P. Georgiou

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1. (a) Sketch the circuit of a VGS multiplier with equations describing its operation.

[4]

(b) Sketch the circuits of two supply independent current sources, including circuitry to guarantee their operation on start-up and explain their main differences.

[6]

(c) Using the equation for fractional temperature coefficient, explain which of your two circuits in 1(b) is more insensitive to temperature.

[5]

(d) Sketch a suitable two transistor and four transistor potential divider voltage reference circuit and show that when VDD=+5V, VSS=-5V and ID=10μA, for an output voltage Vout=0, the four transistor circuit consumes less area than the two transistor one. You may assume K_P=8μA/V^2, K_N=17μA/V^2, V_{TN}=V_{TP}=1V.

[5]

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2. Figure 2 shows a partially-designed two-stage CMOS op-amp required to give a voltage gain of approximately 80 dB, a slew-rate of 3 *V/μs* and a gain-bandwidth product of 3 MHz.

(a) Given that the technology is a fixed 5μm double metal CMOS process, design the channel widths of transistors Q1, Q2 and Q6 for the op-amp to meet the above performance specifications. Aspect ratios of all other devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below.

[12]

(b) Show with equations, and describe why a resistor in series with the compensation capacitor C_C in Figure 2 can significantly improve the amplifier's phase margin.

[4]

(c) Discuss any limitation to the PSRR performance of this circuit and sketch an updated circuit that obtains better PSRR performance.

[4]

CMOS TRANSISTOR MODEL PARAMETERS

MODEL PARAMETERS	Kp ($\mu A/V^2$)	λ (V^{-1})	V_{T0} (V)
PMOS	20	0.03	-0.8
NMOS	30	0.02	1.0

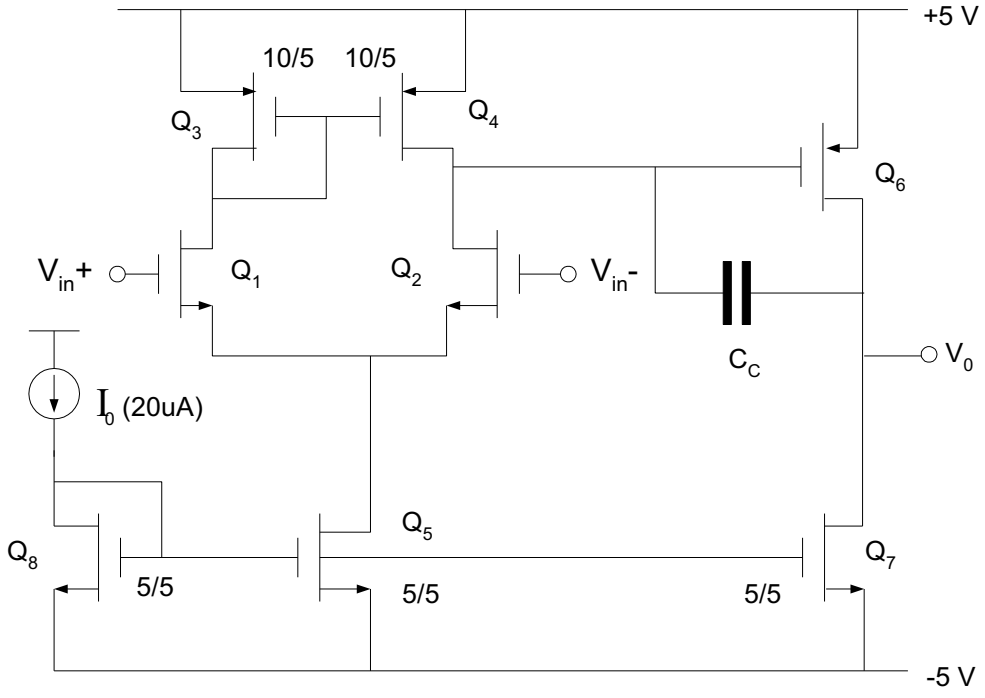


Figure 2

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3. (a) Give one advantage and one disadvantage of each of the two CMOS current mirror circuits shown in Figures 3.1 and 3.2.

[4]

(b) For the current mirror of Figure 3.2 derive this voltage swing in terms of device threshold voltage V_T, clearly stating any assumptions you make.

[6]

(c) (i) Show how a PMOS variant of the current mirror in Figure 3.1 can be used as the active load of a long-tail pair and derive the differential voltage gain.

[4]

(ii) Replace the current mirror to a cascode variant and show the revised voltage gain equation.

[4]

(d) Sketch a regulated cascode current mirror.

[2]

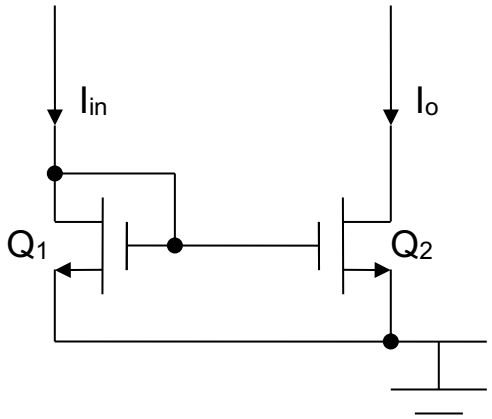


Figure 3.1

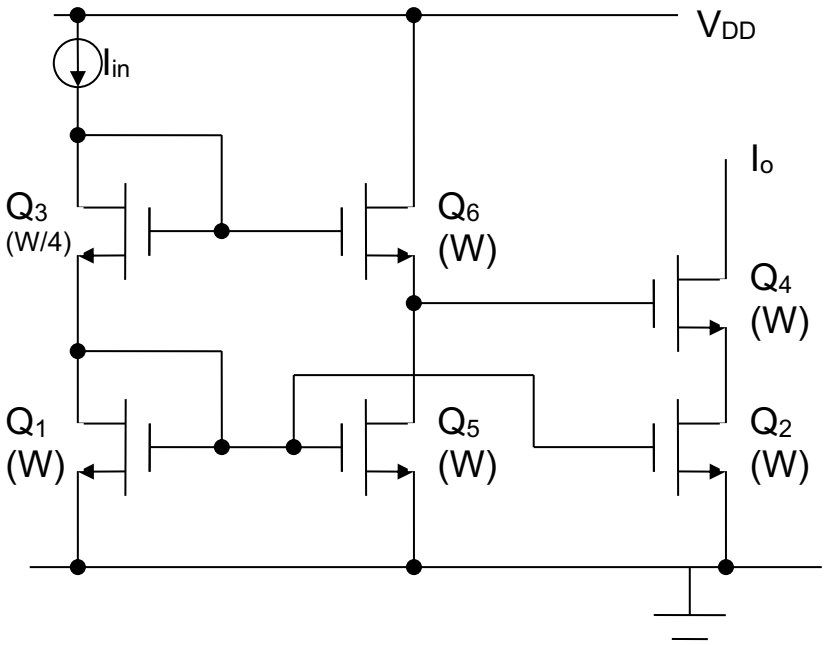


Figure 3.2

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4. (a) Figure 4.1 shows a commonly used circuit for providing a stable current reference. The circuit has the following values I_O = 5μA, R_1 = 14.4KΩ, I_{IN} = 1mA, R_2 = 26.5KΩ. R_2 is a diffused resistor with a temperature coefficient of +1500ppm/°C at T=300K. The output current sensitivity to the supply voltage V_{cc} is given by:

$$S_{V_{CC}}^{Io} = \frac{1}{1 + \frac{I_O R_2}{V_T}} S_{V_{CC}}^{I_{IN}}$$

(i) Derive the term S_{VCC}^{Iin} for the circuit in Figure 4.1 and thus the output current sensitivity.

[5]

(ii) Calculate the change in I_o if V_{CC} where to change by 20%.

[5]

(b) Figure 4.2 shows a differential opamp which is based upon a single-stage fully differential design.

(i) Transistors Q1 and Q2 of Figure 4.2 could be considered to form a linear differential resistor. Derive the equation for linear differential resistance. How does the body effect affect the linearity of each of these transistors?

[5]

(ii) Assuming the time constant of the circuit is τ=10ms, calculate the value of feedback capacitor, C, assuming input transistors Q1 and Q2 are operating in the triode region and have W/L=5, K=20μA/V2, Vg=1V, Vt=0.5V and node x is at a common mode of 0 V. You may neglect any body effects.

[5]

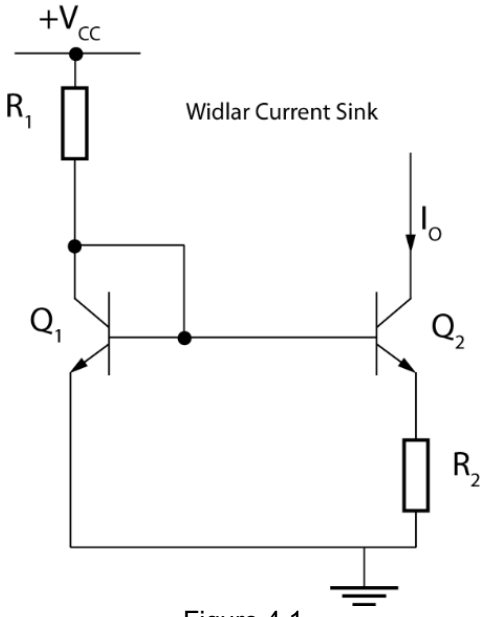


Figure 4.1

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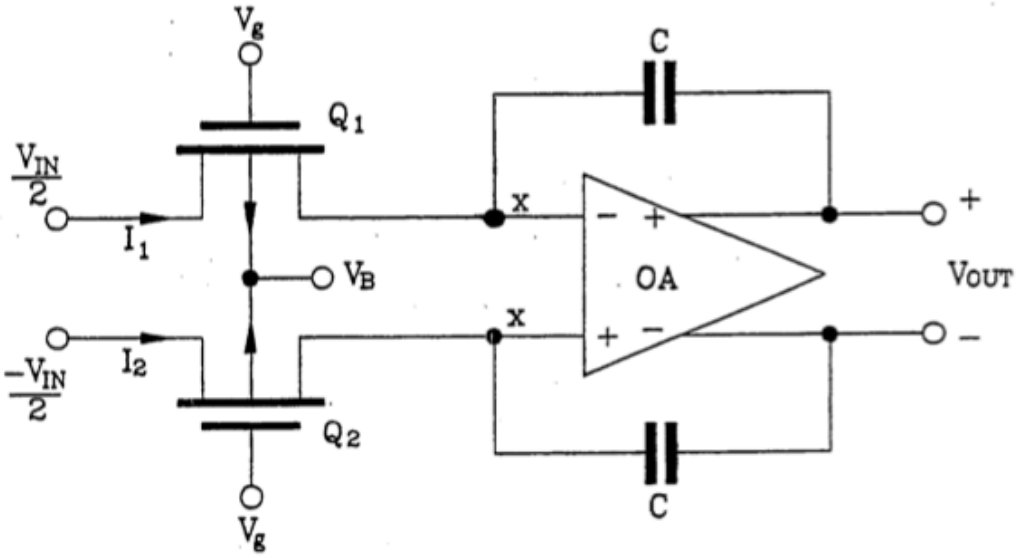


Figure 4.2

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5. (a) Figure 5.1 shows one bit of an algorithmic data converter.

(i) Explain how the data conversion works. [3]

(ii)Show how this can be cascaded to perform an 8-bit data conversion. [4]

(b) What noise limits the resolution of sampled-data converters? [3]

(c) Show how the integrator of Figure 5.2 can be used to mimic both the inductor and capacitor of an LCR passive ladder filter. [10]

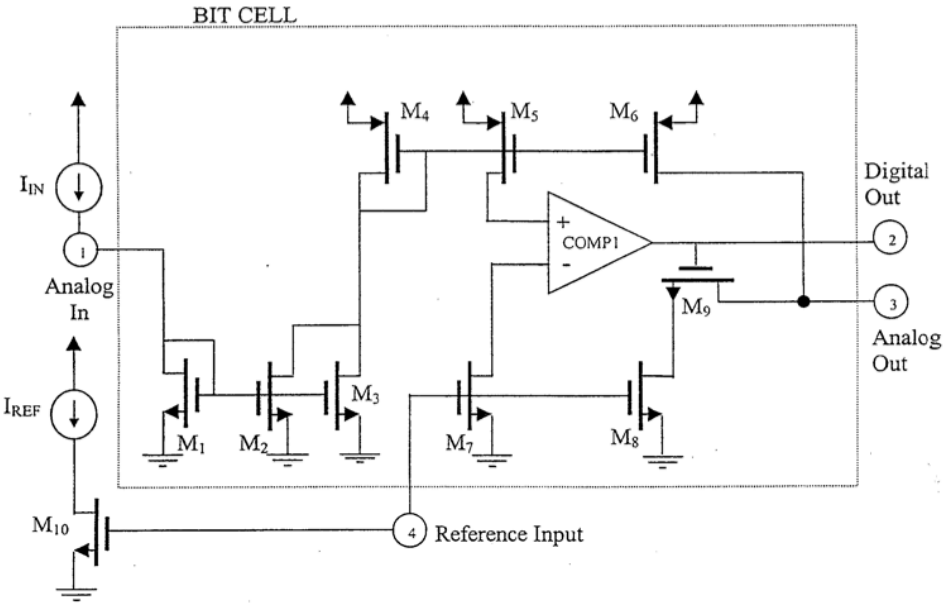


Figure 5.1

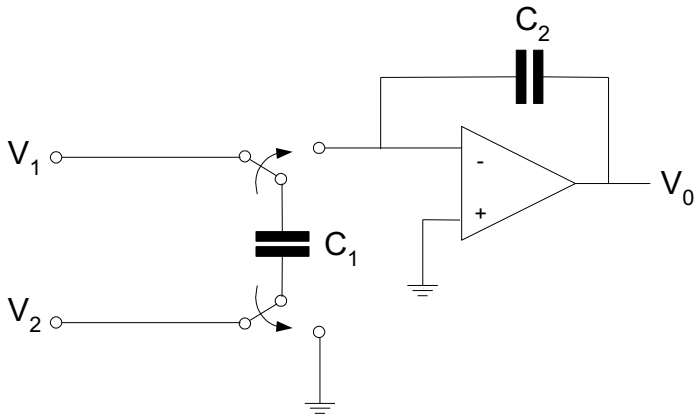


Figure 5.2

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6. (a) Sketch the circuit of a single-stage differential to single ended output high frequency operational amplifier that does not require miller compensation.

[5]

(b) When matching transistors special care is taken with the layout. Discuss what is meant by common centroiding when matching two transistors.

[5]

(c) For fully differential output op-amps explain what is meant by common mode feedback using a diagram where necessary.

[5]

(d) Sketch the circuit of a differential operational amplifier that has a single dominant pole and uses common mode feedback to stabilise the output DC voltage.

[5]

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